Digital Material Passport

ID **TEST-SPECIMEN-001** - *Version* **0.1.0**
Issue Date **2025-09-29** - *Certificate Type*

Manufacturer	Customer	Business Transaction
Steel Testing Labs GmbH 123 Test Street 10115 Berlin DE	**Customer Industries Inc** 456 Customer Ave New York 12345 US	**Order** ORD-2025-001 **Delivery** DEL-2025-001 · **1000 kg**

Product

Product Name	High Strength Steel Plate	*Shape*	Plate - Width 2000 mm - Thickness 50 mm - Length 6000 mm
Batch ID	BATCH-2025-001		

Chemical Analysis

Heat Number **H123456** - *Melting Process* **EAF+LF** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-09-28**

Symbol	*Unit*	*Min*	*Max*	*Actual*
C	%		0.2	0.18
Mn	%	1.4	1.7	1.45
Si	%		0.5	0.35
P	%		0.025	0.012
S	%		0.015	0.008
Cr	%		0.3	0.25

Symbol	*Unit*	*Min*	*Max*	*Actual*
Ni	%		0.2	0.15
Mo	%		0.1	0.08
Cu	%		0.25	0.18
Al	%	0.02	0.05	0.025
N	%		0.012	0.008
CEV	%		0.45	0.42

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength[1] Specimen: 1/4T, L	Rm	485 MPa	450 MPa	600 MPa	ASTM E8	✓
Charpy V-Notch Impact Temperature: -40°C Specimen: 1/4T, L-T	KV	48 / 52 / 45 J - Average 48.3 - Min 45 - Max 52 - Std Dev 3.5	40 J		ASTM E23	✓
Through-Thickness Tensile Room temperature, through-thickness direction					EN 10164	✓

Location (position)	**0**	**0.25**	**0.5**	**0.75**	**1**
Value [MPa]	475	485	490	487	478

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Yield Strength[1] Specimen: Surface, T - ID: YS-SURF-T-001	Re	355 MPa	335 MPa		ASTM E8	✓
Elongation[1] Specimen: Custom (Mid-radius at end section), C - ID: ELONG-MR-001	A	22 %	20 %		ASTM E8	✓

[1] Room temperature

Validation

We certify that the material described herein has been tested and inspected in accordance with the specified standards and meets all requirements.

Validated by **Dr. Hans Schmidt**, Quality Manager - 2025-09-29